DAFFA AHMAD PANGREKSA

As a Data Science undergraduate at Universitas Negeri Surabaya (UNESA), Daffa has developed strong analytical and technical skills through hands-on experience in market research, machine learning, and academic mentoring. Currently serving as a Product Research Specialist Intern at SF Group Enterprise, he conducts competitor analysis, customer sentiment research, and transforms findings into actionable insights to support product development decisions. His experience as a Teaching Assistant has strengthened his technical foundation in Python, algorithms, and computational thinking while enhancing his communication and mentoring abilities. Recognized through multiple achievements in data science and scientific essay competitions, Daffa is passionate about leveraging data-driven approaches to solve real-world problems and contribute to impactful projects in analytics, research, and technology.

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PROFESSIONAL EXPERIENCE

SF Group Services Enterprise Remote
Product Research Specialist Intern May 2026 - Present

- Conducted weekly competitive research across 5 product niches per cycle, analyzing a minimum of 15 competitor product listings on Etsy to identify pricing strategies, market positioning gaps, and untapped product opportunities.
- Performed customer sentiment analysis on competitor product reviews, extracting actionable quality improvement and positioning insights that directly contributed to the launch of multiple new products within the company.
- Delivered 4+ weekly business intelligence presentations to stakeholders using Canva and Excel, coordinating research workflow progress and team task tracking through Trello.
- Accumulated competitive analysis covering 60+ competitor products during onboarding, establishing a recurring benchmarking framework spanning multiple product categories to support data-driven strategy decisions.

Laboratorium Big Data Analysis, FMIPA UNESA Surabaya, East Java
Data Science and AI Researcher Apr 2026 - Jun 2026

- Conducted research in Graph Neural Networks (GNN) for large-scale relational data modeling using Neo4j and NetworkX, targeting faculty research publication and collaboration datasets across Universitas Negeri Surabaya.
- Designed and built a structured experiment pipeline using Jupyter Notebooks to construct, visualize, and analyze academic knowledge graphs and research collaboration networks spanning the full university research ecosystem.
- Developed GNN model architectures to extract meaningful relational patterns from heterogeneous academic graph data, with research outputs targeting peer-reviewed publication in applied graph-based artificial intelligence.
- Built a comprehensive academic graph database integrating co-authorship networks, citation links, and research topic clustering to support institutional research intelligence and cross-department collaboration identification.

Data Science Study Program, FMIPA UNESA Surabaya, East Java
Teaching Assistant in Data Structure and Algorithms Apr 2026 - Jun 2026

- Mentored 70+ students on core algorithmic concepts including search and sorting algorithms, recursion, tree traversals, graph algorithms, and time and space complexity analysis across structured weekly laboratory sessions.
- Designed and graded practical programming assignments with targeted technical feedback on algorithm efficiency, code readability, and correctness, contributing to a consistent class average performance in the B+ to A grade range.
- Reinforced practical comprehension of data structure implementations in Python through guided problem-solving exercises, whiteboard walkthroughs, and individual code review sessions during lab hours.
- Produced supplementary reference materials and worked example sets for complex topics including dynamic programming and graph traversal, improving student comprehension across algorithmic problem types.

Data Science Study Program, FMIPA UNESA Surabaya, East Java
Teaching Assistant in Basic Programming Apr 2026 - Jun 2026

- Mentored 70+ students on foundational programming concepts including variables, control flow, functions, file I/O, and introductory object-oriented programming across structured weekly laboratory sessions.

- Evaluated programming assignments and delivered targeted technical feedback on code logic, syntax conventions, and problem decomposition, supporting a consistent class performance average in the A- to A grade range.
- Facilitated hands-on problem-solving exercises in Python, guiding students through debugging workflows and algorithmic thinking approaches to build independent coding competency from the ground up.
- Supported individual student development through one-on-one consultation sessions during lab hours, addressing misconceptions in fundamental programming logic and reinforcing best practices in structured code design.

PROJECT EXPERIENCE

Machine Learning Project - National Essay Competition by CSSMORA ITS Surabaya, East Java
RADENSA: Rainfall-Dengue Spatiotemporal Forecasting System May 2026 - Jun 2026

- Developed Indonesia's first weekly sub-district dengue forecasting system for 44 DKI Jakarta kecamatan, integrating 20,592 multi-source observations to transition from reactive to proactive surveillance.
- Engineered 22 climate-epidemiological predictors and benchmarked various models via spatiotemporal cross-validation to ensure high predictive accuracy.
- Deployed a high-performance LightGBM dashboard (R^2: 0.77, MAE: 2.95) to support real-time surveillance and Indonesia's Zero Death Dengue 2030 target.
- Conducted spatiotemporal analysis integrating multi-source administrative and epidemiological data to capture geographically heterogeneous dengue transmission dynamics across Jakarta's urban sub-district landscape.
- Secured 2nd Place nationally at CSSMORA ITS among 120 competing submissions, demonstrating the system's scientific rigor and real-world applicability as a high-impact public health decision support tool.

Statistical Modeling Project Surabaya, East Java
CB-SEM Analysis on E-Book Reader Adoption Determinants Apr 2026 - Jun 2026

- Conducted CB-SEM on 174 respondents using an Extended Technology Acceptance Model (TAM) with 6 latent constructs to analyze e-book reader adoption intention.
- Validated model assumptions with Mardia's Test and applied MLR estimation with Yuan-Bentler correction to address non-normal response distributions.
- Resolved discriminant validity failure between Perceived Usefulness and Compatibility (HTMT = 0.961) by merging both into a unified Functional Value construct.
- Performed convergent validity, VIF multicollinearity diagnostics, and BCa bootstrapping, ensuring rigorous psychometric validation throughout the SEM workflow.
- Identified Emotional Value as the strongest predictor of adoption intention ($\beta = 0.521$, $p < 0.001$), with the final model explaining 58.4% of variance ($R^2 = 0.584$).

Data Mining & Deep Learning Project Surabaya, East Java
TweetTrade: Twitter Sentiment-Driven Stock Price Prediction for BBCA Apr 2026 - Jun 2026

- Built an end-to-end deep learning pipeline predicting BBCA daily closing prices using Indonesian Twitter sentiment, OHLCV, and macroeconomic data (2019–2026), processing 18,217 tweets via IndoBERT for sentiment extraction.
- Generated 5 sentiment metrics and applied Genetic Algorithm feature selection with Ridge regression to reduce 12 technical indicators to the 8 most predictive features.
- Implemented 5-fold walk-forward cross-validation across 22 variable configurations and optimized hyperparameters with Optuna, benchmarking LSTM, GRU, BiLSTM, and Attention-LSTM architectures.
- Achieved best performance with LSTM on KomVar8 (IHSG + sentiment features) at RMSE 232.72 IDR, MAPE 2.60%, and R^2 0.9189; sentiment features significantly improved accuracy ($p < 0.0001$).
- Validated trading utility through backtesting, delivering +2.19% alpha over the Buy-and-Hold benchmark, with model predictions interpreted using SHAP analysis.

Data Engineering Project Surabaya, East Java
TalentTrack: Automated Data Warehouse and NLP Pipeline for Job Market Intelligence Apr 2026 - Jun 2026

- Designed an automated ETL pipeline using Apache Airflow to process 124,000+ Kaggle job records and daily JobSpy extractions, loading data into a PostgreSQL star-schema warehouse.
- Applied GLiNER zero-shot NER to extract 36 skill labels from job descriptions, combined with NLTK language detection and regex fallback for multilingual postings.

- Architected an OLAP star schema with 5 dimension tables, optimizing query performance using pg_trgm, btree_gin indexes, and materialized views for real-time dashboards.
- Automated skill demand forecasting using Prophet and Holt-Winters models, enabling monitoring of emerging technology trends from daily job market data.
- Processed 7,000+ active postings and identified 9,267 unique job-skill relationships, revealing Game Development roles command the highest salary premiums despite lower posting volumes than Management roles.

Machine Learning Project Surabaya, East Java
Machine Learning for Job Market Intelligence: NER, Clustering, and Salary Prediction Apr 2026 - Jun 2026

- Designed a three-task machine learning pipeline for LinkedIn job market analysis covering skill extraction, semantic job clustering, and log-transformed salary prediction from real-world job postings.
- Fine-tuned JobBERT for NER to extract certifications, technical skills, and tools from job descriptions using manually annotated labels, enabling scalable structured skill extraction.
- Applied SBERT embeddings with UMAP reduction and benchmarked HDBSCAN against GMM, with HDBSCAN producing 37 job clusters at Silhouette Score 0.4419.
- Built a 604-dimensional feature matrix combining SBERT embeddings, TF-IDF SVD representations, and job structure features, comparing XGBoost, LightGBM, and Stacking Ensemble models.
- Deployed a Streamlit salary prediction application using XGBoost, while the offline Stacking Ensemble achieved the best performance with R^2 0.6603.

Computer Vision Project Surabaya, East Java
Metal Casting Defect Detection: Robustness Analysis and Unsupervised Localization Apr 2026 - Jun 2026

- Developed a metal casting defect inspection framework using 8,126-dimensional HOG and LBP features, evaluating SVM robustness across 5 sensor degradation and domain-shift scenarios.
- Simulated industrial degradation through Gaussian noise ($\sigma = 25$) and zoom-crop distortions, revealing accuracy collapse from 99.16% on clean data to 60.84% under noisy conditions.
- Identified Scenario 5b (denoising + zoom-crop training augmentation) as the most robust configuration, maintaining 76.00% accuracy under spatial distortion while improving generalization across degraded inputs.
- Designed a denoising pipeline combining Bilateral Filter and Non-Local Means, restoring classification accuracy to 98.60% by preserving HOG feature integrity.
- Trained a Convolutional Autoencoder for unsupervised defect localization, achieving 100% recall across 200 defective samples with an average of 5.91 bounding boxes per image.

Machine Learning Project - National Essay Competition by KSE USU Surabaya, East Java
GeoStunting - Geospatial ML for Stunting Risk Prediction Apr 2026 - May 2026

- Developed a geospatial machine learning framework for stunting-risk prediction and mapping across 514 districts and cities in Indonesia, supporting precision targeting and resource allocation for the MBG Program.
- Integrated socioeconomic, nutritional, and geographic datasets at district level to engineer spatially aware features capturing vulnerability patterns, health infrastructure access, and demographic risk factors across regions.
- Built a Hybrid Geographically Weighted Regression and Random Forest model with a Composite Risk Score approach, improving prediction performance from baseline R^2 0.36 to R^2 0.62 through model tuning.
- Translated prediction outputs into prioritized intervention maps to facilitate evidence-based targeting of MBG disbursement across high-risk districts, directly applicable to national stunting reduction policymaking.
- Secured 1st place nationally among 56 teams at KSE Juara 2026 Scientific Essay Competition, with all 8 finalists representing distinct universities, validating the framework's scientific rigor and policy relevance.

Deep Learning Project - National Infographic Competition by FILM UNS Surabaya, East Java
SiPangan: Deep Learning Early Warning System for Rice Price Surge Prediction Apr 2026 - May 2026

- Developed a deep learning early warning system predicting national rice price surges 1–3 months in advance, using 196 months of historical data from 2010–2026 to support smart farming and post-harvest sales optimization.
- Applied Pearson and Spearman Correlation with Mutual Information to select 37 informative features from 46 variables, including ENSO Index, FAO Food Price Index, USD/IDR exchange rate, and Bulog rice stock levels.
- Built a two-layer LSTM with a 6-month sliding window for price regression and a SMOTE-balanced Random Forest Classifier for surge classification, outperforming the ARIMA baseline across all evaluation metrics.
- Achieved R^2 0.7179, MAE Rp167/kg, and MAPE 1.21% on the LSTM regression task, generating Smart Farming and Post-Harvest Signals to guide optimal planting and selling decisions for Indonesian farmers.

- Placed Top 15 nationally as Semi-Finalist at the Semar Infographic Competition among 100+ submissions, with projected impact reducing national food loss exposure by Rp551 Trillion annually.

Data Wrangling Project Surabaya, East Java
East Java Regional Food Affordability Index Nov 2025 - Dec 2025

- Integrated datasets from UMK wage documents, Bank Indonesia price bulletins, and Tokopedia scraping across 11 food commodities and 9 East Java cities into a unified analysis-ready dataset.
- Developed a Tokopedia web scraping pipeline to supplement offline market benchmarks with real-time e-commerce prices, enabling city-level food affordability comparisons.
- Built two composite indices, Market Affordability Index (MAI) and E-Commerce Access Index (EAI), identifying a 2× purchasing power gap between Surabaya and Sumenep.
- Validated the indices through correlation analysis and benchmarking against provincial wage statistics, revealing structural inequalities in regional food access.
- Published the curated dataset on Kaggle to support regional economic research, food security modeling, and evidence-based policymaking in East Java.

Inferential Statistics Project Surabaya, East Java
Benzene Estimator: Multi-Sensor Air Quality Analysis Nov 2025 - Dec 2025

- Developed a Streamlit web application for real-time benzene (C_6H_6) concentration prediction using multi-sensor data from the UCI Machine Learning Repository.
- Built an OLS regression pipeline with automated IQR-based outlier removal, evaluating performance using R^2, MAE, and RMSE metrics.
- Implemented interactive Plotly visualizations for residual analysis, feature contributions, and cross-sensor correlation heatmaps within the web interface.
- Performed feature engineering on 9 metal oxide sensor readings with humidity and temperature variables to identify the most predictive sensor combinations.
- Analyzed sensor collinearity using correlation matrices and Variance Inflation Factor diagnostics to improve model reliability and interpretability.

Artificial Intelligence Project Surabaya, East Java
Diabetes Risk Prediction: NHANES 2017–2020 Dataset Sep 2025 - Dec 2025

- Integrated 10 CDC NHANES 2017–2020 XPT datasets, consolidating demographic, anthropometric, and clinical variables for diabetes and prediabetes risk stratification.
- Engineered composite biomarkers including Lipid Accumulation Product (LAP), Visceral Adiposity Index (VAI), and a custom Metabolic Risk Score to capture latent metabolic risk factors.
- Implemented data leakage detection and correction protocols, removing temporally inappropriate variables and retraining models under leakage-free conditions.
- Benchmarked XGBoost, Random Forest, and MLP classifiers using stratified cross-validation and threshold tuning, with Random Forest achieving F1 0.526 and AUC 0.895.
- Applied SMOTE oversampling and threshold optimization to improve minority-class recall for prediabetes detection and early-risk screening.

Text Processing Project Surabaya, East Java
Comparative Sentiment Analysis of the MBG Program Using LDA and IndoBERT Sep 2025 - Dec 2025

- Collected and processed 8,386 social media records comprising 4,968 X/Twitter posts and 3,418 YouTube comments on the MBG Free Nutritious Meal Program using a standardized NLP preprocessing pipeline.
- Fine-tuned IndoBERT for Indonesian sentiment classification, achieving an average inference confidence of 0.89 across informal policy-related social media discourse.
- Applied LDA topic modeling to identify latent themes and compare topic distributions between X/Twitter and YouTube communities discussing MBG.
- Conducted cross-platform sentiment analysis, finding higher negative sentiment on YouTube (49.1%) than X/Twitter (46.9%), indicating stronger criticism in long-form discussions.
- Synthesized findings into a comparative public perception report, providing actionable insights for government communication and program evaluation.

Digital Signal Processing Project Surabaya, East Java
Music Genre Classification and Recommendation System Oct 2025 - Nov 2025

- Designed a digital signal processing pipeline on the 999-track GTZAN dataset, applying a fifth-order Butterworth band-pass filter and extracting 58 features including MFCC, ZCR, Chroma STFT, spectral centroid, and rolloff.
- Engineered a feature extraction workflow spanning temporal, spectral, and tonal domains, with normalized representations for fair cross-model benchmarking.
- Benchmarked MLP, SVM, KNN, and Decision Tree classifiers for 10-class music genre classification, with MLP achieving the best performance at 76.50% accuracy and 0.767 F1 score.
- Analyzed confusion matrices to identify systematic misclassification between acoustically similar genres, informing targeted feature refinement.
- Built a cosine similarity-based music recommendation engine using acoustic feature proximity to enable cross-genre discovery and personalized recommendations.

UI/UX Project Surabaya, East Java
TrashCycle: Sustainable Waste Management Mobile Application Sep 2025 - Nov 2025

- Designed a waste management mobile application prototype comprising 40+ screens and user flows using Figma under a user-centered design framework.
- Conducted persona development, user journey mapping, and wireframing activities to identify key pain points in waste sorting and recycling participation.
- Developed an integrated Waste Bank ecosystem featuring waste categorization, scheduled pickups, reward points, voucher redemption, and recycled product marketplace modules.
- Created high-fidelity prototypes covering end-to-end waste management workflows, including collection requests, financial incentives, and sustainability tracking.
- Evaluated usability through user feedback sessions, where 88.3% of respondents assigned an overall experience score of 8/10 or higher.

Business Intelligence Project Surabaya, East Java
Superstore Sales & Inventory Analytics Dashboard Sep 2025 - Nov 2025

- Developed an end-to-end pipeline using Pandas to integrate transactional, customer, product, and inventory data, processing 5,009 orders with total revenue of $2.39M and profit of $299K.
- Built an interactive Streamlit dashboard featuring dynamic filtering across date ranges, regions, and product categories, supported by 10+ business intelligence visualizations.
- Designed KPI monitoring modules tracking revenue, profit, profit margin (12.5%), customer segment contribution, monthly sales trends, and top-performing states.
- Implemented correlation and profitability analyses across product categories and sub-categories, identifying revenue drivers and loss-generating product segments.
- Developed an automated inventory alert system detecting products with critical stock levels (<20 units), enabling proactive inventory monitoring and replenishment planning.

Data Mining Project - DISCO Competition Surabaya, East Java
MSME Revenue Prediction - Data Mining Competition (DISCO) May 2025 - Jun 2025

- Built an MSME profit prediction model using XGBoost with Optuna hyperparameter optimization, achieving R^2 0.91 on 13,564 business records across retail, service, and manufacturing sectors.
- Conducted feature engineering and selection to identify key profit drivers, including transaction frequency, average order value, and customer retention metrics.
- Developed percentile-based business segmentation classifying MSMEs into low-, medium-, and high-profitability tiers, providing structured performance benchmarking across sectors.
- Validated model robustness through k-fold cross-validation and residual analysis, ensuring consistent generalization and minimizing sector-specific overfitting.
- Placed 6th among 20+ teams in the national DISCO Data Mining Competition organized by Universitas Trunojoyo Madura, demonstrating strong predictive modeling and analytical capabilities.

ORGANIZATIONAL EXPERIENCE

ACTION 2025: National Academic Competition of Data Science - HIMASADA UNESA Surabaya, East Java
Event Division Staff Jun 2025 - Dec 2025

- Managed the national infographic competition track at ACTION 2025, attracting 74 submissions from 150+ participants nationwide and generating IDR 2.59 million in registration revenue.
- Developed competition guidelines, evaluation rubrics, and submission validation procedures to ensure consistent and transparent judging standards.
- Moderated judging sessions involving 3 expert judges, facilitating structured deliberation and accountable scoring processes.
- Compiled and validated submission records, scoring matrices, and finalist rankings, ensuring a smooth and contestation-free award announcement.
- Collaborated with cross-functional committees to coordinate judging schedules, participant communications, and end-to-end competition execution.

RELOUD Inaguration 2024 - HIMASADA UNESA Surabaya, East Java
Logistic and Supplies Division Staff Jun 2025 - Dec 2025

- Managed end-to-end logistics for an inauguration event with 300+ participants, achieving 100% equipment readiness and zero operational incidents.
- Coordinated procurement, inventory, and equipment distribution across 5+ internal divisions and external vendors, delivering approximately 15% cost savings within budget.
- Developed logistics timelines and equipment checklists to identify supply gaps early and ensure seamless event execution without delays.
- Liaised with venue, catering, and audio-visual partners to coordinate setup, operations, and breakdown while maintaining full operational continuity.

EDUCATION

Universitas Negeri Surabaya Surabaya, East Java
Bachelor of Data Science. GPA 3.92 / 4.0 Aug 2024 - Present
Relevant Courses: Python, R, Figma, Microsoft Office, Project Management, Statistics, Machine Learning, Data Analysis, Data Scientist, Data Engineer, Database, PostgreSQL, Docker, Dashboard Data Visualization, Data Mining

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SKILLS & LANGUANGES

Technical Skills: Python (Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Statsmodels, Librosa), R, Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision (CV), Graph Neural Networks (GNN), Data Analyst, Data Scientist, Data Engineer, Apache Airflow, Docker, PostgreSQL, MySQL, Database Development, Statistics, Mathematics, Power BI, Tableau, Microsoft Office, Figma, Canva, UI/UX Development, UI Design, UX Research, Competitive Analysis, IT Management, Computer Architecture, Competition Management, Laboratory Assistance

Soft Skills: Analytical Skills, Problem Solving, Attention to Detail, Teamwork, Team Management, Project Management, Competitive Intelligence, Financial Skills, Entrepreneurship, Personal Development, Higher Education

Language: Indonesian (Native), English (Limited working proficiency)

ACHIEVEMENTS & AWARDS

- 1st Place – KSE Juara 2026 National Scientific Essay Competition (Data Science Track), organized by Paguyuban KSE Universitas Sumatera Utara, 2026
- 2nd Place – CSSMORA ITS National Scientific Essay Competition, organized by CSSMORA ITS, 2026
- Top 15 National Semi-Finalist – Semar Infographic Competition, organized by FILM UNS, 2026
- 6th Place – DISCO Data Mining Competition, organized by Universitas Trunojoyo Madura, 2025

- 9th Place – National Mathematics Olympiad (GISN), organized by PT Digital Edu Indonesia and Indonesia Youth Festival (IYFEST), 2025

CERTIFICATION

- Data Science Certification – Dicoding Indonesia
- Fundamentals of Artificial Intelligence Certification – Dicoding Indonesia
- Financial Literacy Certification – Dicoding Indonesia
- Data Analyst Certification – Digital Talent Scholarship (Komdigi/Kemenkominfo)